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10 CONCEPTUAL Q&A DATA PLATFORMS · PRACTICE WORKSHEET

CONCEPTUAL SERIES

Amazon Redshift

Name: ________ Cohort/Batch: ________ Date: ________ Score: ________

AMAZON REDSHIFT PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Data Platforms

TOTAL: 10 QUESTIONS

Q1 What is Amazon Redshift and what problem in Data Platforms does it solve? *Threat Model*

Q2 What are the core components or concepts in Amazon Redshift? *Architecture*

Q3 How does Amazon Redshift compare to its main alternatives? *Configuration*

Q4 How do you get started with Amazon Redshift for a new project? *Hardening*

Q5 What configuration options most affect Amazon Redshift's behavior in production? *SSO / IdP*

Q6 How does Amazon Redshift handle reliability and high availability? *Observability*

Q7 What observability does Amazon Redshift provide out of the box? *Lifecycle*

Q8 What are common performance bottlenecks in Amazon Redshift? *Compliance*

Q9 What security best practices apply when running Amazon Redshift? *Testing*

Q10 What mistakes do teams commonly make when adopting Amazon Redshift? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Amazon Redshift is a tool or platform

Mitigation

Amazon Redshift is a tool or platform that automates or simplifies a specific concern in the Data Platforms domain, reducing manual effort and operational complexity for engineering teams.

A2 Amazon Redshift has key building blocks that

Primitives

Amazon Redshift has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A3 Amazon Redshift trades off simplicity against feature

Hardening

Amazon Redshift trades off simplicity against feature richness differently than its alternatives. Choose Amazon Redshift when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A4 Install Amazon Redshift via its package manager

Gotchas

Install Amazon Redshift via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A5 Production-critical settings typically include concurrency

Federation

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A6 Amazon Redshift provides HA through leader election,

Observability

Amazon Redshift provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A7 Amazon Redshift exposes metrics (Prometheus endpoint or

Automation

Amazon Redshift exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A8 Performance bottlenecks typically stem from misconfigured

Governance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A9 Run Amazon Redshift with the principle of

Assessment

Run Amazon Redshift with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A10 Common pitfalls include skipping the documentation and

Optimization

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.